

SAM (SHENGKAI) XU

Ph.D. Researcher | Machine Learning, Computer Vision & Engineering Systems

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EXPERIENCE

Research Assistant
UNC Charlotte, Dr. Hongfei Xue
2024 – Present

- Architect, train, and evaluate deep-learning and computer-vision models for multimodal sensing with mmWave radar, RGB/depth cameras, LiDAR, and acoustic sensors.
- Build ground-truth and sensor-data pipelines for pose and mesh reconstruction; analyze accuracy, latency, and cross-modal consistency and communicate results to cross-disciplinary collaborators.

AI Research Assistant
UNC Charlotte TeCSAR Lab
2022 – 2023

- Developed a video-processing and anomaly-action-detection pipeline for civilian security and public-safety applications.
- Increased processing throughput by 50%, improving video smoothness and system response time.

Embedded Firmware Engineer
Oxit LLC
2021 – 2023

- Developed modular Embedded C libraries across chipsets and built low-power RF and IoT systems.
- Designed AWS and Google Cloud automation pipelines and tools that improved development efficiency and compatibility.

SELECTED PROJECTS

Neural Musculoskeletal Pose Estimation
Human Mesh Reconstruction, Inverse Kinematics | 2026–Present

- Real-time mesh and pose pipeline being translated toward deployment with collaborators in sports medicine.

SocialBit On-Device Classifier
Wearable ML, Audio Classification | 2026–Present

- Privacy-focused smartwatch models for interaction type, conversational depth, and tone under edge-compute constraints.

SELECTED PUBLICATIONS

Real-Time Neural Musculoskeletal Pose Estimation
IEEE/ACM CHASE 2026

- Replaced iterative inverse kinematics with a neural surrogate in a real-time monocular reconstruction pipeline.

Towards Robust mmWave-Based Human Activity Recognition Using Large Simulated Datasets for Model Pretraining
IEEE BigData 2024

- Used simulated human-mesh data and heatmap perturbations to improve pretraining with limited real-world data.

Real-Time Object Detection Algorithm Performance on Edge Computing Devices
IEEE ISNCC 2024

- Benchmarked YOLOv5 latency and power on Jetson Nano, Raspberry Pi 4, and Coral TPU.

PROFILE

Ph.D. researcher developing deep-learning models and experimental pipelines for physical sensing, edge deployment, and engineering prediction. Embedded-systems and CAD experience; Formula SAE contributor motivated to apply AI and scientific computing to motorsports.

EDUCATION

Ph.D. Computer Science
UNC Charlotte
Expected 2029

Research focus: deep-learning systems for multi-modal sensing.

M.Sc. Computer Engineering
University of North Carolina at Charlotte
Aug 2022 – Aug 2024

Research focus: machine learning on edge and embedded systems.

B.Sc. Computer Engineering
University of North Carolina at Charlotte
Jan 2018 – Dec 2022

Machine Learning concentration; Mathematics minor

TECHNICAL SKILLS

Programming & ML
Python, C++, Embedded C, SQL, PyTorch, TensorFlow, OpenCV, scikit-learn

Engineering & Systems
SOLIDWORKS, KiCad, sensor integration, experimental data acquisition, embedded systems, Linux, Git/GitHub, AWS, Google Cloud

MOTORSPORTS INVOLVEMENT

Formula SAE Electrical Team | 2018–2019
Contributed to electrical power distribution and vehicle wiring for UNC Charlotte Formula SAE.

Club-Level Automobile Time Attack | 2020–2024
Participated in time-attack events and prepared personal track cars through mechanical maintenance, repairs, inspections, and track-day setup.

Motorcycle Track Days | 2024–Present
Participate in track days and prepare personal motorcycles through routine service, mechanical repairs, and pre-event safety inspections.